

Figure 2: QXL device and driver

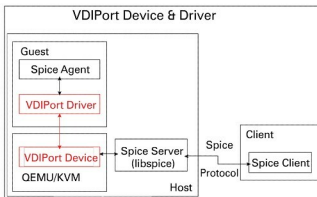


Figure 3: VDI port device and driver

The SPICE server interacts with the remote client (SPICE client) on the outside, while on the inside it interacts with the virtual devices that are exposed to it by QEMU running over KVM. Figure 1 depicts SPICE architecture.

The desktop being accessed from the remote location is actually a virtualised guest that exposes its devices—typically, display, audio and USB—to the SPICE server to connect with. Thus, SPICE will only work when the remote hosts are virtual machines. The remotely connecting client connects to the SPICE server using the SPICE protocol, which communicates events like keyboard and mouse inputs, graphics commands, audio streams, etc.

The advantage of the above is an OS-independent protocol that only needs the SPICE client. You can install the SPICE client and run it using a Web browser like Firefox. Moreover, SPICE is able to achieve great performance when compared to other environments because a virtual GPU (called QXL) has been developed for SPICE to improve its graphical

performance. This supports active display on the remote client from the boot stage—something not possible in traditional SBC environments, since the remote hosts to be connected to have to remain switched on and booted. Red Hat has released QXL PCI drivers for Windows and Linux, though standard VGA is supported if no drivers are available.

The graphics rendering subsystem differs from traditional SBC environments in another significant manner. Most SBC environments send graphical updates to the remote client by updating frame buffers. SPICE supports transmission and handling of 2D graphics commands. SPICE uses Cairo, a 2D graphics, device-independent library and Pixmap for manipulating pixel regions (www.cairographics.org), apart from using image compression, video compression and caching to increase the client side display rendering experience.

Another important component in the SPICE scheme of things is the SPICE agent. This resides on the virtualised guest and supports a communication channel between the SPICE client and the agent. Yet another virtual device, the VDI Port—a QEMU PCI device—is used for this communication. This virtual device is used by the client to communicate with the agent. For a virtualised Windows guest, the agent is a Windows service that runs on the guest.

SPICE also has audio and USB support. At a recent demonstration, we used two different clients to connect to the same guest. We played a song on one of the clients (a Linux desktop), logged out of it and connected through a laptop (running Windows) to hear the same song being played. The customer has since evinced a keen interest in deploying SPICE for his office over a WAN link. The demonstration elicited the same “Wow” that I had experienced when I first came across this wonderful technology—a simple protocol that can really add some spice to the growing desktop virtualisation market. Feel free to contact me, in case you have any specific queries about SPICE or its implementation. **END** 

By: Varad Gupta

Varad is an open source enthusiast who strongly believes in the open source collaborative model not only for technology but also for business. India's first RHCSS (Red Hat Certified Security Specialist), he has been involved in spreading open source through Keen & Able Computers Pvt Ltd, an open source systems integration company, and FOSTERING Linux, a FOSS training, education and research training centre. The author can be contacted at varad.gupta@fosteringlinux.com

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